Higher National Diploma in Engineering (Civil/Mechanical)

Higher National Diploma in Building Services Engineering

First Year, 2nd Semester Examination-2021

ME 1206/BSE 1205 - Fundamentals of Thermodynamics

answer any five questions.

For air, R = 0.287kJ/kgK, C_P = 1.005kJ/kgK

Time : 3 hours

Question 01

(i) State the **first law of thermodynamics**. (02 marks) (ii) State the **Non flow energy equation**. (02 marks) (iii) Explain in brief, what is "**internal energy**" of a system. (04 marks) (iv) A $15\ \mathrm{cm}$ diameter vertical cylinder, closed by a piston contains a combustible mixture at a temperature of $30^\circ C$. The piston is free to move and its weight is such that the mixture pressure is $3\ \mathrm{bar}$. Upper surface of the piston is exposed to the atmosphere. The mixture is ignited. As the reaction proceeds, the piston moves slowly upwards and heat transfer to the surroundings takes place. When the reaction is complete and the contents have been reduced to the initial temperature of $30^\circ C$, it is found that the piston has moved upwards a distance of $8.5\ \mathrm{cm}$ and the magnitude of heat transfer is $4\ \mathrm{kJ}$. (12 marks) (a) Sketch the $\mathrm{p\text{-}V}$ diagram for the process. Calculate: (b) Work done. (c) Change in internal energy of the system.

Question 02

(i) What is an **adiabatic process**? (02 marks) (ii) A closed system undergoes an adiabatic compression process from state 1 to state 2. Obtain a relationship between temperatures T_1, T_2 and volumes V_1, V_2. The index of compression is γ. (04 marks) (iii) A certain quantity of air has a volume of $0.028\ \mathrm{m}^3$ at a pressure of $1.25\ \mathrm{bar}$ and $25^\circ C$. It is compressed to a volume of $0.0042\ \mathrm{m}^3$ according to the law $pv^{1.3} = \mathrm{constant}$. (14 marks) (a) Draw the $\mathrm{p\text{-}v}$ diagram for the above processes. Find: (b) Final temperature. (c) Work done during the compression. (d) Change in internal energy. (e) Heat transfer during the process.

Question 03

(i) Briefly explain the following terms. (08 marks) (a) **Dryness fraction**. (b) **Dry saturated steam** . (c) **Superheated steam**. (d) **Critical point**. (ii) Steam engine obtains steam from a boiler through a pipe line. The pressure and dryness fraction of steam produced by the boiler is $15\ \mathrm{bar}$ and 0.98 respectively. It was observed that the steam loses $21\ \mathrm{kJ}$ of heat per kg as it flows through the pipeline, pressure remaining constant. Calculate the dryness fraction of the steam, at the engine end of the pipe line. (08 marks) (iii) A boiler produces steam at $8\ \mathrm{bar}$ pressure and $325^{\circ}C$ temperature. Find the **specific volume** and **specific enthalpy** of the steam produced by the boiler. (04 marks)

Question 04

(i) State the **steady flow energy equation**. (04 marks) (ii) Simplify the steady flow energy equation for the following devices. Clearly state all assumptions made. (06 marks) (a) Compressor. (b) Nozzle. (iii) A steam turbine receives steam of $1.35\ \mathrm{kg/s}$ and the power output is $500\ \mathrm{kW}$. The heat loss from the casing is $125\ \mathrm{kW}$. Calculate the change in **specific enthalpy of steam** across the turbine when the velocities at entrance is $60\ \mathrm{m/s}$, the velocity at exit is $360\ \mathrm{m/s}$, and the inlet pipe is $3\ \mathrm{m}$ above the exhaust line. (06 marks) (iv) Superheated steam at $6\ \mathrm{bar}$ pressure and $400^{\circ}C$ enters a nozzle at velocity of $75\ \mathrm{m/s}$. The velocity of steam at the exit of the nozzle is $350\ \mathrm{m/s}$. Find the **enthalpy of steam** at the nozzle exit. (04 marks)

Question 05

(i) State the **Second Law of Thermodynamics**. (03 marks) (ii) Define **Coefficient of Performance (COP)** for: (04 marks) (a) Refrigerator. (b) Heat Pump. (iii) Show that the COP of a refrigerator is equal to $\frac{1}{\frac{T_H}{T_C}-1}$, where T_c and T_h are cold and hot reservoir temperatures respectively. (04 marks) (iv) A refrigerator extracts heat at a rate of $120000\ \mathrm{kJ/hr}$ from a cold reservoir which is maintained at $-10^{\circ}C$ when power input is $0.75\ \mathrm{kW}$. (09 marks) Find: (a) Rate of heat transfer to the surroundings. (b) COP of a refrigerator. (c) Temperature of surroundings.

Question 06

(i) State four **air standard assumptions**. (04 marks) (ii) A hot air engine works on an air standard cycle which consists of following processes: (16 marks) * 1-2 Constant pressure heating. * 2-3 Adiabatic compression. * 3-4 Constant pressure cooling. * 4-1 Adiabatic expansion.

The initial and final pressures of air are $3\ \mathrm{bar}$ and $1\ \mathrm{bar}$ respectively. The temperatures of air at the beginning of compression and expansion processes are $298\ \mathrm{K}$ and $923\ \mathrm{K}$ respectively. Determine: (a) Temperature of air at the end of compression. (b) Heat supplied per kg of air. (c) Temperature of air at the end of expansion. (d) Work done per kg of air. (e) Thermal efficiency of the engine.